

Array is a DS:

finite ordered set of homogeneous elements.

↳ no of elements are fixed, predefined.

e.g. 2 elements defined earlier.

- Every element is in sequence, denoted by index. (ordered set).

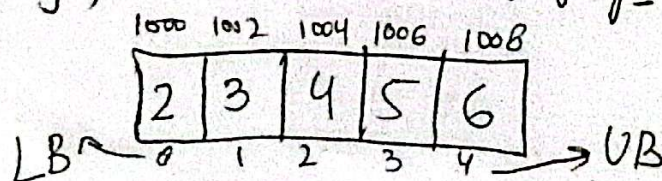
- All the elements of same type, homogeneous elements.

- Elements of array stored in contiguous memory location. (Same memory location, same memory sequence).

- If 5 elements in array, we want to access 4th element, we can randomly (direct) access any element.

int A [5];

Length
Size of Array = $UB - LB + 1$



One Dim, ~~Two~~ Multi Dim Array

Multi Dim. - Array of Array

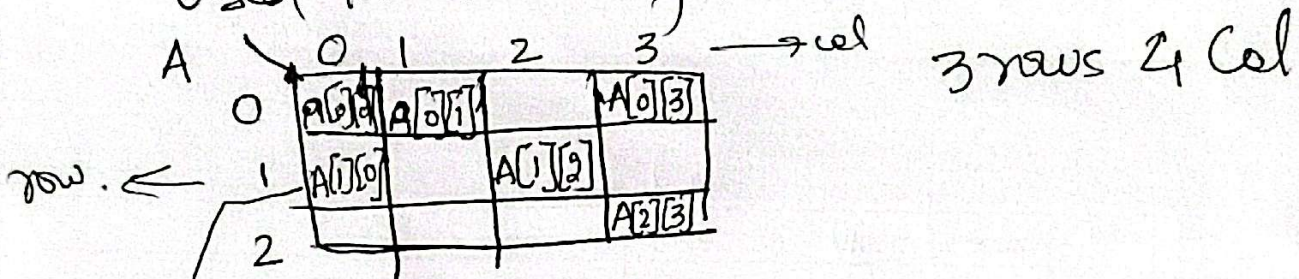
1D Array to access second element, $A[1]$.

multi Dim. One Dim for Row.
" " for Col.

2D Array define as `int a[3][4];`

used two index for row & Column.

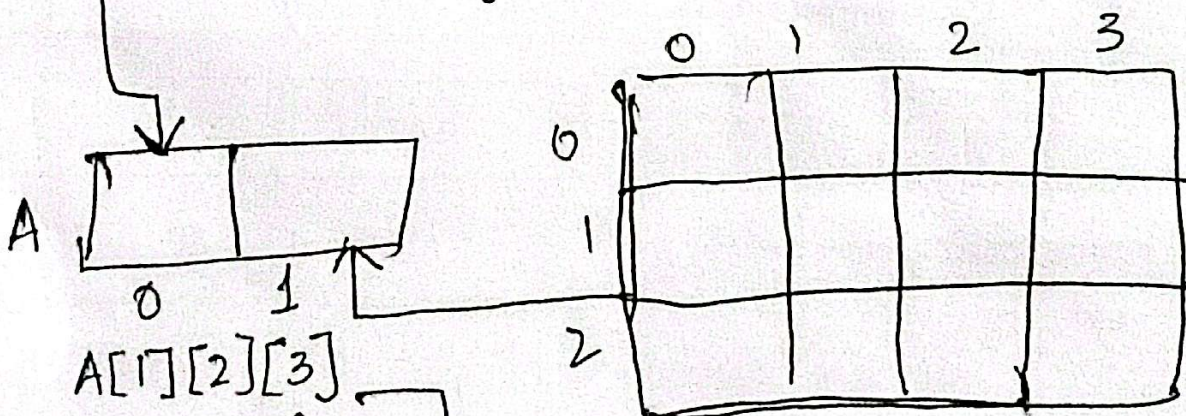
total elements
 $3 \times 4 = 12$



3D Array `int A[2][3][4];`

no. of Arrays ↓ rows ↓ col's.

$2 \times 3 \times 4$
 $= 24$
elements

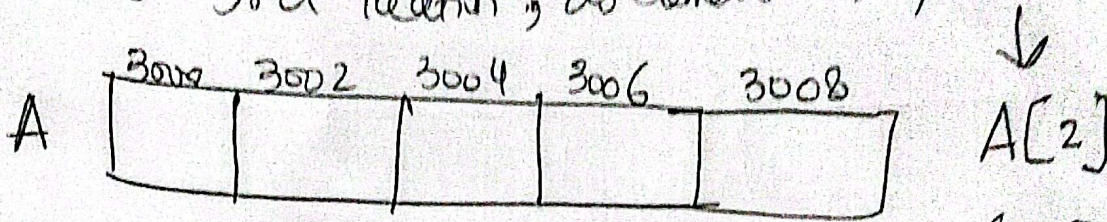


Base Address.

→ Second Array 2nd row, 3rd col

24
24

We can randomly access the array, to direct access 3rd location, as elements in sequence



To find address of $A[2] = 3000 + 2 \times 2$
 Start add. (Base Add.) = 3004

size of 1 element
 no. of elements cross.
 at (2), we cross (0), [1].

$$A[i] = BA + (i - LB) \times S$$

Base address size of each element.

address of ith element in one D Array.

$$A[i] = BA + (i - LB) \times S$$

BA = Base add
 LB = Lower Bound
 S = size

$$[-3 \ 4] \rightarrow LB = -3, \ UB = 4.$$

$$[0 \ 4] \rightarrow LB = 0, \ UB = 4.$$